

# Sara Arif

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## Summary

Computer Science engineer and systems builder focusing on applied machine learning, data engineering, and full-stack software architecture. Experienced in orchestrating multi-agent LLM systems (LangGraph), deploying production web applications (PHP/React), and implementing low-level compiler infrastructure (AST parsing). Proven track record of architecting end-to-end systems that extract, process, and serve complex data securely.

## Education

**University of Chakwal**  
*Bachelor of Science in Computer Science (CGPA: 3.85 / 4.00)*

Expected Sept 2027  
Chakwal, Pakistan

- Relevant Coursework: Data Structures & Algorithms, Compiler Construction, Artificial Intelligence, Operating Systems, Databases.

## Experience

**DevelopersHub Corporation**  
*Data Science Intern (Remote)*

Jun 2025 - Jul 2025  
Lahore, Pakistan

- Engineered custom features and handled severe class imbalances utilizing SMOTE to prepare data for ML models.
- Trained and evaluated predictive models (CatBoost), optimizing specifically for high recall to minimize false negatives.
- Focused on model interpretability (SHAP), ensuring technical outputs could be clearly explained to non-technical stakeholders.

## Technical Skills

**Programming Languages:** Python, C++, C#, SQL (MySQL, PostgreSQL), R, JavaScript, TypeScript, PHP

**AI & Machine Learning:** LangGraph, RAG, HuggingFace, Scikit-learn, CatBoost, SMOTE, TensorFlow

**Data Engineering & BI:** ETL Pipelines, Advanced DAX, Power BI, Tableau, Pandas, NumPy

**Software Architecture:** RESTful APIs, Microservices, Progressive Web Apps (PWA), Relational DB Design

**Tools & Frameworks:** React, FastAPI, Streamlit, Git, Docker, SQLite, Vercel, Figma

## Projects Portfolio

**DatSage Analytics Engine | LangGraph, FastAPI, Streamlit**

AI Agent & Data Workflow

Jan 2026

- Decoupled heavy LLM inference logic into an asynchronous FastAPI backend to prevent client-side UI blocking.
- Managed secure execution sandboxes for dynamically generated Python code to prevent environment corruption.

**Financial Risk Classifier | Python, CatBoost, SMOTE**

Machine Learning Pipeline

Nov 2025

- Applied SMOTE to resolve a 90/10 class imbalance in a credit default dataset.
- Trained a CatBoost classifier and tuned hyperparameters to maximize recall and minimize false negatives.

**DG Cement Work Tracker | PHP, MySQL, React**

Full-Stack Progressive Web App

Aug 2025

- Designed a normalized MySQL schema and PHP REST API to manage inventory logs and shift assignments.
- Deployed a mobile-first interface replacing paper-based tracking, eliminating data entry duplication for supervisors.

**UrduPy Compiler | Python, AST**

Language Parser & Infrastructure

Mar 2025

- Wrote a custom lexical analyzer and recursive descent parser from scratch to process Urdu programming syntax.

- Generated Abstract Syntax Trees (ASTs) to map right-to-left tokens into functionally executable Python logic.

## Certifications & Achievements

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**Silver Honour (Top 2%):** International Computer Science Competition (ICSC) - 3,800+ Participants (2025)

**Generative AI Professional:** Snowflake (2025)

**AI Agents (LangChain & RAG):** DeepLearning.AI (2025)

**Data Analysis with Power BI:** Microsoft (2025)

**Deep Learning with Keras & TensorFlow:** IBM (2025)